

Name:

BUID:

Workshop 3: LU Factorization

1. Determine the matrix that implements the transformation $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ 2x_2 \\ x_3 \end{bmatrix}$.
2. Determine the elementary matrix in $\mathbb{R}^{3 \times 3}$ that implements the row operations $R_3 \leftarrow R_3 - 2R_1$.
3. Determine the elementary matrix in $\mathbb{R}^{4 \times 4}$ that implements the row operations $R_4 \leftrightarrow R_2$.
4. Determine the matrix in $\mathbb{R}^{3 \times 3}$ that implements the following sequence of row operations.

$$R_2 \leftarrow 3R_2$$

$$R_1 \leftarrow R_1 - 2R_2$$

$$R_2 \leftrightarrow R_3$$

5. Let E denote the matrix you got from the previous part and let A denote the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Determine EA by matrix multiplication. Also apply the row operations from the previous problem to A . Verify that you get the same thing.
6. Let A denote the matrix $\begin{bmatrix} 2 & 1 & 2 & 5 \\ 4 & 2 & 8 & 7 \\ -2 & -1 & 2 & -1 \end{bmatrix}$ and let U denote the matrix $\begin{bmatrix} 2 & 1 & 2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 7 \end{bmatrix}$ which is an echelon form of A .
 - (a) Determine the sequence of row operations used in forward elimination (using only replacements) to transform A to U .
 - (b) Determine the matrix that implements the row operations from part (a).
 - (c) Determine the sequence of row operations that “inverts” the operations from part (a), i.e., that transforms U to A .
 - (d) Determine the inverse of the matrix from part (c).
 - (e) Determine an LU -factorization of A .
 - (f) Describe the relationship between the row operations in part (a) and the entries of the matrix L from part (d).

Write your solutions below and on the back on this page.

Continue your work here.